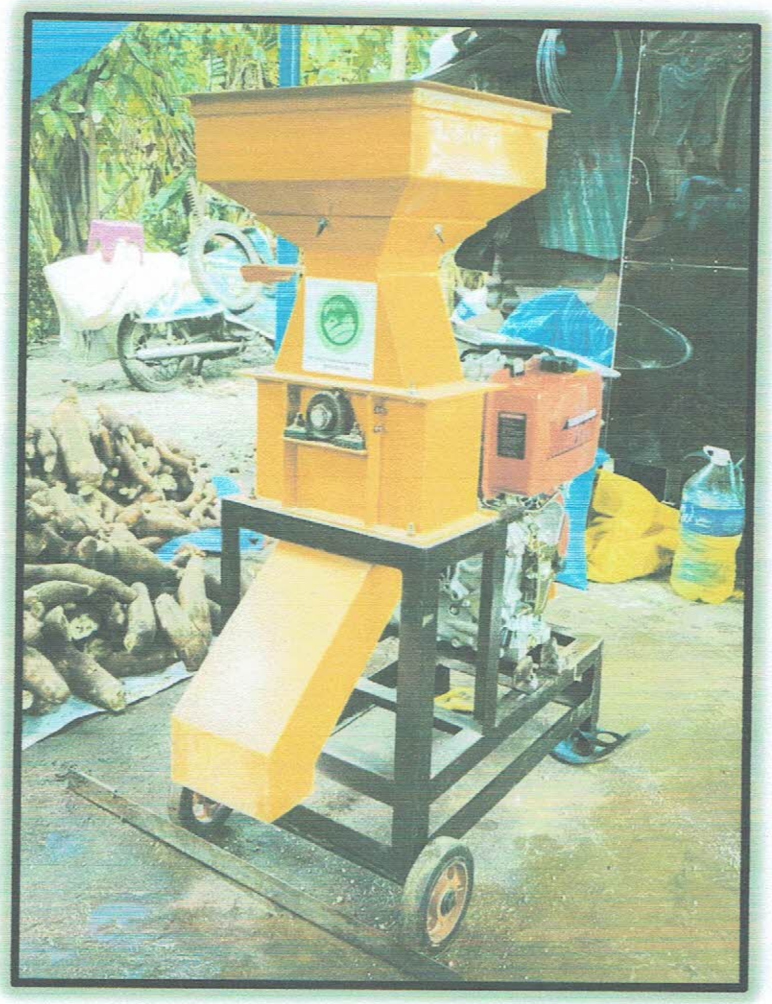
Test Report No. 2023 – 0013



Agricultural Machinery – Hammer Mill
ACPF HAMM2

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Test Report Number	2023 – 0013
Date of Issuance	January 30, 2023
Valid Until	January 30, 2028
Agricultural Machinery	Hammer Mill
Type	Swinging, Engine-driven
Brand	ACPF
Model	HAMM2
Test Requested by	ACPF AGRICULTURAL MACHINERY TRADING San Juan Libmanan, Camarines Sur

SUMMARY

Performance Criteria	Requirement as per PAES 216:2004	AMTEC Test
Fineness Modulus	Course Grinding: 4.80 (4.20 – 5.40)[a]	4.31
Noise Level, dB(A), maximum	96[b]	100.0

[a] Refer to Table 1 of PAES 216:2000; fineness modulus for shelled corn, coarse grind.
[b] Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Not provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	ACPF AGRICULTURAL MACHINERY TRADING
4. Prime mover used	8.95 kW YAMASAKI BYK188FA Diesel Engine
5. Number of operators	Two
6. Loading of input	Input materials were manually loaded into the input hopper.
7. Cleaning of parts	Parts can be accessed for cleaning by removing the covers.
8. Adjusting and repairing of parts	Parts are locally fabricated and can be repaired. The feeding rate can be adjusted through a sliding gate provided.
9. Collecting of output	Output is collected using a basin.
10. Transporting the machine	The machine is equipped with transport wheels for easy transport.
11. Safety features	None
12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	No failure or abnormalities observed on the machine or its component during and after operation.
13. Steel bars and heavy-duty mild steel shall be generally used for the manufacture of the different components of the hammer mill.	The machine is generally made up of steel bars, GI sheet and cold rolled steel.
14. Hammers shall be made of AISI 1080 to AISI 1085 or its ISO equivalent.	Not verified

OBSERVATIONS (Cont'n)

15. Bolts and screws to be used shall conform with the requirements of PAES 311 and PAES 313.	Not verified
16. Hammers and perforated screen shall be accessible and replaceable.	The hammers and perforated screen can be accessed and replaced by opening the milling chamber.
17. Hubs and spacing collars shall have matching diameters and faces to afford a tight fit and to eliminate the formation of shoulders.	Diameters and faces of hubs and spacing collars were not measured, hence matching was not verified.
18. All components shall be dynamically balanced for stable running with low noise levels.	No excessive vibrations of the machine observed during the test operation.
19. There shall be provision to prevent metallic materials from entering the milling chamber.	There was no provision that will prevent metallic materials from entering the milling chamber.
20. Sealed type bearings shall be used in the product zone.	Sealed type bearings were used in the product zone.
21. The hammer mill shall be free from manufacturing defects that may be detrimental to its operation.	The hammer mill is free from any manufacturing defects that may be detrimental to its operation.
22. Any uncoated metallic surfaces shall be free from rust and shall be painted properly.	All metallic surfaces were painted except for the milling assembly.
23. The hammer mill shall be free from sharp edges and surfaces that may injure the operator.	The hammer mill is free from sharp edges that may injure the operator.
24. Belt cover or guard and provisions for belt tightening and adjustments shall be provided.	Belt cover or guard was not provided.
25. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 19, 2023
Location of Test	Brgy. Binobong, Pili, Camarines Sur 13° 31' 54.30" N, 123° 15' 37.73" E
Items Tested	Operator's Manual, Speed of Components, Noise Level, Input Capacity, Fuel Consumption Rate, Output Fineness Modulus, and Number of Operators

METHODS OF TEST

The hammer mill was tested based on PAES 217:2004 Agricultural Machinery – Hammer Mill – Methods of Test.

DESCRIPTION OF THE MACHINE

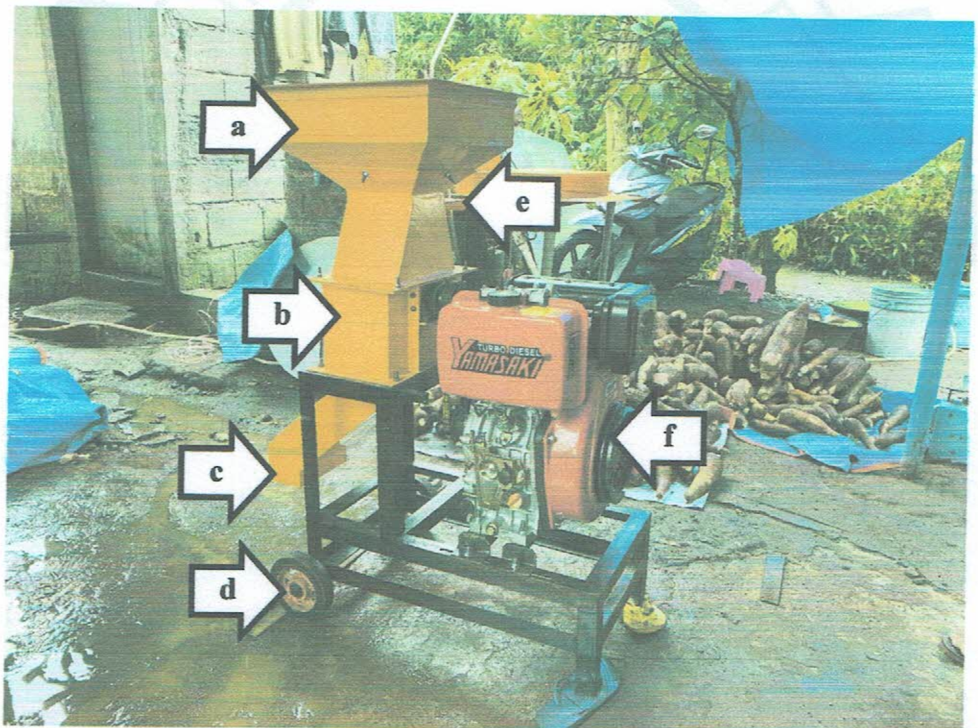

Figure 1. Parts of the ACPF HAMM2 Hammer Mill powered by 8.95 kW YAMASAKI BYK188FA Diesel Engine include the following: (a) Input hopper, (b) Milling chamber, (c) Output chute, (d) Transport wheels, (e) Feed intake adjustment, and (f) Prime mover.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	800	860
1.1.2	Width	400	501
1.1.3	Height	1100	1100
1.2	Weight without primemover, kg	ND	NM
2	Intake hopper		
2.1	Dimensions, L × W × H, mm	410 × 410 × 230	355 × 357 × 214
2.2	Inlet dimensions, L × W, mm	130× 150	127 × 148
2.3	Height from the ground, mm	950	1100
2.4	Material	G.I Sheet	G.I. Sheet
3	Milling assembly		
3.1	Type	ND	Hammermill, swinging
3.2	Dimensions, W × D, mm	ND	150 × 243
3.3	Number of anchor bar	8	8
3.4	Hammer		
3.4.1	Dimensions, L × W × t, mm	NM	52.70 × 23.75 × 5.11
3.4.2	Number of blades per anchor bar	6 and 8	6-8-6-8-6-8-6-8
3.4.3	Means of attachment	ND	Slotted
3.4.4	Material	Cold rolled steel	ND
4	Screen		
4.1	Dimensions, L × W, mm	ND	NM
4.2	Number of screens	ND	1
4.3	Mesh number	ND	ND
4.4	Perforation, D, mm	ND	5.02
4.5	Material	ND	ND
4.6	Clearance between screen and tip of the hammer, mm	ND	25
5	Aspirator	NA	NA

[a] The data were obtained from the specifications sheet provided by the requesting party.
Note: NA – Not Applicable, ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
6	Engine		
6.1	Brand	YAMASAKI	YAMASAKI
6.2	Model	BYJ188FA	BYK188FA
6.3	Type	Diesel engine	Single vertical cylinder, air-cooled, diesel engine
6.4	Make or manufacturer	ND	China
6.5	Serial number	ND	BTY20150420016
6.6	Rated power, kW	8.95	NM
6.7	Rated speed, rpm	3600	NM
6.8	Displacement, cm^3	0.418	NM
6.9	Fuel type	Diesel	Diesel
6.10	Fuel tank capacity, L	ND	NM
6.11	Cooling system	Air-cooled	Air-cooled
6.12	Starting system	Recoil starter	Rope-recoil starter
7	Power transmission system		
7.1	Engine to milling assembly	ND	V-belt and pulley
7.1.1	Engine[b]	ND	75 × 1B × 64
7.1.2	Milling assembly[b]	ND	75 × 1B × 64
7.1.3	Belt size	B-28	B-28
8	Safety devices	None	None
9	Special features	None	None

[a] The data were obtained from the specifications sheet provided by the requesting party.
[b] Pulley diameter, mm × number of belts × shaft diameter, mm
Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Conditions of test material		
1.1	Crop	Shelled corn	
1.2	Source	Pili, Camarines Sur	
1.3	Variety	White corn (treated for planting)	
1.4	Moisture content, $\%_{wb}$	13.25	
1.5	Purity, %	99.28	
1.6	Bulk density, kg/m^3	725.11	
1.7	Dimensions, mm		
1.7.1	Length	9.14	
1.7.2	Width	8.15	
1.7.3	Thickness	2.80	
2	Performance test		
2.1	Speed of components, rpm	Without load	With load
2.1.1	Engine shaft	2527	2427
2.1.2	Hammer mill shaft	2529	2397
2.2	Noise level, dB(A)	Without load	With load
2.2.1	Feeding operator's ear level	97.6	98.0
2.2.2	Bagger's ear level	98.7	100.0
2.3	Fuel consumption rate, L/h	0.787	
2.4	Weight of input, kg	31.12	
2.5	Weight of output, kg	30.68	
2.6	Milling capacity, kg/h	210.10	
2.7	Input capacity, kg/h	221.92	
2.8	Output capacity, kg/h	210.91	
2.9	Product recovery, %	98.60	

RESULTS (Cont'n)

Table 1. Laboratory Analysis of Milled Corn

Standard US Sieve Size No.	Nominal Sieve Opening, mm	Material Retained on Each Sieve, %
3/8	9.42	0.00
4	4.76	0.00
8	2.38	55.89
16	1.19	28.07
30	0.595	9.55
50	0.297	3.78
100	0.149	2.60
Pan	0.037	0.110
Fineness modulus		4.31
Grinding classification		Coarse
Average particle size, mm		2.08

OBSERVATION

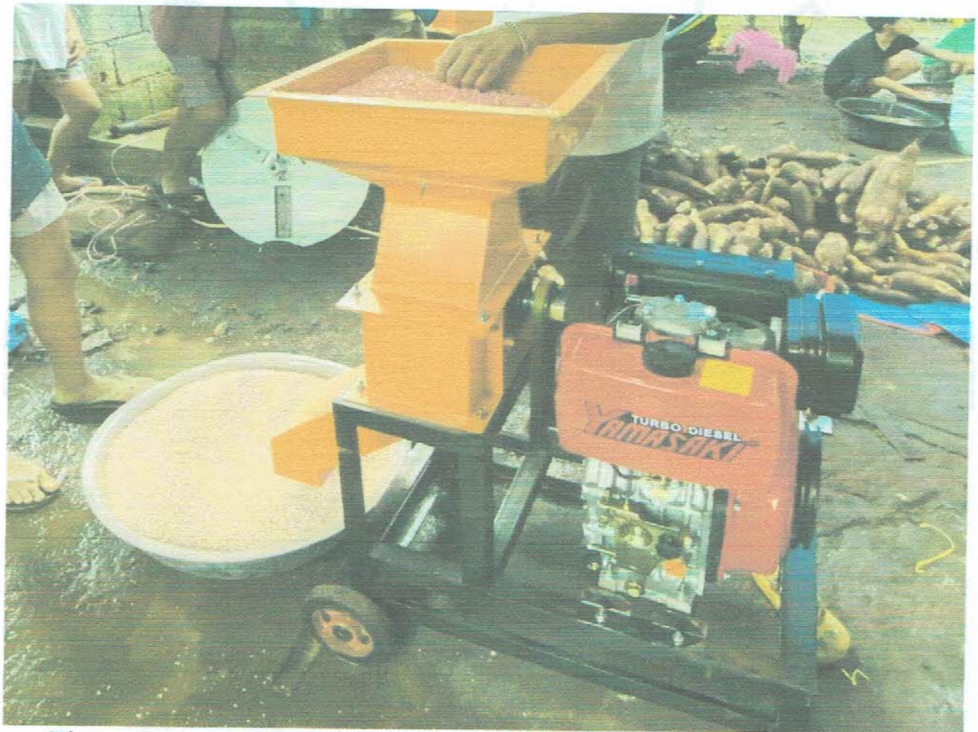

Figure 2. Actual performance test of ACPF HAMM2 Hammer mill.

Tested by:

MARK ANTHONY R. ISUREÑA
Test Engineer

Verified by:

FATIMA JOY J. RAYTANA
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO	FEB 01 2023
AMTEC Director	Date

REMINDER

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